

LIGHTWEIGHT SAFE HELICOPTER

LSH 00 — COMPREHENSIVE ENGINEERING BASELINE

Generation 1 | Engineering-Strengthened Preliminary Controlled System Baseline

Document ID	LSH 00
System Name	Lightweight Safe Helicopter
Variant	Generation 1 — Separable Survival Capsule Concept
Revision	3.2 — Strengthened Engineering Baseline
Status	[PROPOSED] Preliminary Controlled Baseline
Founder	Amirbahador Atai
Language	English
Certification status	Not a certification approval; no flight or performance claim is demonstrated
Primary use	Root system baseline for requirements, architecture, analysis, V&V, certification and licensing documentation

Engineering integrity statement: LSH 00 is a system-level preliminary baseline. It deliberately does not present unverified aircraft performance, emergency-separation capability, crashworthiness, parachute performance, reliability or certification compliance as facts. Any item that requires analysis, test, inspection or certification evidence remains explicitly open. All mass, power, timing and performance figures are planning values only and carry no demonstrated status.

Revision 3.2 strengthening principle: *Safety functions, independent initiation paths, physical separation functions and post-event protection are distinguished rather than treated as automatically independent layers. Candidate jettison items are not pre-credited as safely releasable. Rotor loading terminology is corrected and unambiguous. Power targets are retained only as study ranges, never as feasibility evidence. Parachute count is determined by sizing and reliability analysis. Partial and asymmetric separation, common-cause failures and downstream state authorisation are treated as first-class design cases. Requirements, targets, assumptions, calculations and evidence remain strictly status-tagged and traceable.*

1. PURPOSE, SCOPE AND BASELINE BOUNDARY

- Define the system-level baseline for a lightweight twin-engine helicopter concept incorporating an occupied survival capsule and a controlled emergency separation / jettison function.
- Provide the root structure from which a 400–500 document engineering tree can be decomposed without silently converting assumptions into requirements.
- Define requirements, targets, assumptions, calculated quantities, verification gates, interfaces, hazards and configuration-control rules.
- Provide a common technical language for engineering, verification, certification, manufacturing and technology-licensing work.

Boundary: LSH 00 is not a detailed design, production drawing package, certification approval, flight manual or demonstrated performance report. No statement in this document constitutes a claim of feasibility, airworthiness or operational safety.

2. SYSTEM DESCRIPTION

- Conventional twin-turboshaft, single-main-rotor helicopter with tail rotor and fixed skid or short fixed-support landing gear.
- Nominal occupancy: 7 persons (two flight crew + five additional occupants).
- A dedicated survival capsule forms the final occupied landing boundary after any successful emergency separation.
- The emergency architecture is intended to reduce the mass and fire hazard carried into the capsule's post-separation landing phase. Separation shall receive survival credit only after the relevant verification evidence exists.
- Normal flight, emergency separation, parachute descent and ground impact are treated as coupled system states, not as independent demonstrations.

3. STATUS AND CONFIGURATION LEGEND

Tag	Meaning	Use
[REQ]	Allocated requirement	Baseline obligation unless changed by formal configuration control
[TARGET]	Design target	Objective subject to engineering closure; not a demonstrated value
[ASSUMP]	Engineering assumption	Temporary analytical basis with identified owner and closure plan
[CALC]	Calculated quantity	Valid only for the stated inputs, equations and configuration
[OPEN]	Open verification / analysis	No compliance or performance claim permitted
[TBD]	Not yet defined	Must be resolved by a controlled downstream document
[DEMO]	Demonstrated	Only after objective evidence is formally accepted

4. TOP-LEVEL SYSTEM REQUIREMENTS

ID	Requirement / Target	Status
LSH-REQ-SYS-001	The aircraft shall provide seating and restraint provisions for 7 occupants.	[REQ]
LSH-REQ-SYS-002	The Generation 1 architecture shall use two engines and one main rotor with a tail rotor.	[REQ]
LSH-REQ-SYS-003	The occupied survival capsule shall constitute the defined final landing boundary.	[REQ]
LSH-REQ-CAP-001	The capsule shall include occupant protection, impact attenuation, recovery and post-event survival provisions.	[REQ]
LSH-REQ-JET-001	Emergency separation shall have at least one automatic initiation path and an independent manual crew initiation path; any backup automatic path shall be independently assessed.	[REQ]
LSH-REQ-JET-002	Unintended separation / jettison shall be treated as a catastrophic hazard and shall receive a quantitative safety objective in the applicable certification basis.	[REQ]
LSH-REQ-JET-003	No component shall be designated as universally jettisonable until structural, dynamic, fire, control and collision analyses show that release is acceptable for the applicable envelope.	[REQ]
LSH-TGT-MASS-001	15 000 kg is the Generation 1 planning MTOW for system studies only.	[TARGET]
LSH-TGT-CAP-001	4 500 kg is the Generation 1 planning complete-capsule mass.	[TARGET]
LSH-TGT-JET-001	≤ 6 s from validated emergency command to verified capsule-clear state is a development objective only.	[TARGET]
LSH-TGT-ROTOR-001	15.0 m is the main-rotor planning diameter.	[TARGET]
LSH-TGT-RANGE-001	750–900 km is a preliminary range objective, subject to aerodynamic, propulsion, mass and fuel closure.	[TARGET]
LSH-TGT-END-001	45–55 min is a preliminary hover-endurance objective, subject to power, fuel and thermal closure.	[TARGET]

5. MASS PROPERTIES AND CONFIGURATION STATES

The 15 000 kg value is an accounting baseline, not a verified aircraft mass. The mass model must close for all safety-critical configurations, including normal flight, fuel extremes, emergency separation, capsule-only descent, asymmetric release and failed-release cases.

Item	Planning mass	Status
Complete survival capsule	4 500 kg	[TARGET]
Main airframe excluding capsule	3 800 kg	[ASSUMP]
Main rotor + hub + controls	1 450 kg	[ASSUMP]
Tail rotor + drive	280 kg	[ASSUMP]
Engines + gearboxes + mounts	1 350 kg	[ASSUMP]
Fuel system + usable fuel	1 650 kg	[ASSUMP]
Rapid-separation hardware + independent hardware	650 kg	[ASSUMP]
Avionics / electrical / hydraulic / misc.	1 070 kg	[ASSUMP]

Item	Planning mass	Status
Mass growth allowance	250 kg	[TARGET]
ACCOUNTING TOTAL	15 000 kg	[CALC]

- Mass properties shall include CG envelope, inertia tensor, uncertainty / measurement allowance and configuration-dependent mass properties.
- Fuel mass shall be reconciled using the selected fuel specification and tank geometry; the stated 1 650 kg allocation shall not be interpreted as a validated fuel volume or energy content.
- Capsule mass shall be decomposed bottom-up before it becomes a controlled design requirement.
- Mass growth allowance is a program reserve, not evidence that the design has demonstrated margin.

6. GEOMETRY AND ROTOR LOADING DEFINITIONS

Parameter	Planning value	Status
Capsule length	5.20 m	[TARGET]
Capsule width	2.45 m	[TARGET]
Capsule height	1.95 m	[TARGET]
Aircraft overall length	15.5–16.2 m	[TARGET]
Fuselage width	2.6–2.8 m	[TARGET]
Overall height	4.2–4.5 m	[TARGET]
Main rotor diameter	15.0 m	[TARGET]
Tail rotor diameter	2.7 m	[TARGET]

Rotor terminology correction: For $D = 15.0$ m, disk area $A = \pi D^2/4 \approx 176.715$ m². At the 15 000 kg planning mass, conventional weight-based disk loading is approximately 84.88 kg/m². A separate thrust margin or thrust-to-weight quantity must never be labelled “disk loading”. Any additional ratio retained by later studies shall be defined explicitly by operating condition, thrust definition and power state.

7. PROPULSION, ROTOR AND PERFORMANCE CLOSURE

- Two turboshaft engines remain the Generation 1 architectural requirement.
- No specific engine power is locked by LSH 00. A previous 1 150–1 300 shp per-engine range is retained only as a candidate study envelope and shall not be used as evidence of feasibility.
- Power-required versus power-available analysis shall cover gross mass, density altitude, temperature, rotor speed, hover, climb, cruise, autorotation-related states, transient limits and one-engine-inoperative (OEI) conditions.
- Transmission, gearbox, shafting, rotor aerodynamic performance, thermal margins and engine-control limitations shall be included.
- Range and endurance shall be derived from a closed mass, drag, rotor, engine, fuel and mission model.

Closure item	Minimum evidence	Status
Power matching	Validated analytical model with defined margins and operating points	[OPEN]
Rotor performance	Aerodynamic model / correlation, figure of merit and operating envelope	[OPEN]
Transmission	Power, torque, thermal and transient margins	[OPEN]
OEI	Defined OEI requirement and compliance analysis	[OPEN]
Range / endurance	Mission model with fuel, reserves and environmental conditions	[OPEN]

8. SURVIVAL CAPSULE ARCHITECTURE

Element	Baseline definition	Status
Primary structure	CFRP roll-cage with metallic / titanium nodes and controlled joints	[TARGET/REQ]
Occupancy	7 occupants	[REQ]
Seats / restraints	Energy-absorbing seats with occupant restraint system	[REQ]

Element	Baseline definition	Status
Landing attenuation	Progressive mechanical energy-absorption system; final geometry TBD	[REQ/TBD]
Parachute system	Distributed multi-canopy architecture; count / configuration determined by sizing	[REQ/TBD]
Normal-flight attachment	Defined primary load path with controlled shock isolation	[REQ]
Emergency release	Dedicated controlled release path with positive retention	[REQ]
Post-event systems	Recovery aids and required life-support provisions retained with capsule	[REQ]

Important design principle: The capsule's ability to protect occupants after a failed or partial separation is not credited merely because the capsule is structurally strong. It must be demonstrated for the actual governing impact, parachute, load-path and occupant-load cases.

9. MULTI-LAYER SAFETY ARCHITECTURE

LSH uses a layered safety architecture, but the layers are not assumed to be statistically independent. The architecture separates four different ideas: hazard detection, command initiation, physical separation, and residual occupant protection. Independence is a verification claim, not a naming convention.

Layer	Safety function	What it protects against	Independence / limitation
S1	Emergency detection / automatic initiation	Failure of crew recognition or delayed response	Dedicated sensing / logic path; common-cause analysis required
S2	Independent backup initiation	Failure of S1	Separate sensing / processing / power where practicable; independence not credited until analysed
S3	Manual crew initiation	Automatic detection / logic failure	Direct crew path with guarded controls and clear state feedback
S4	Controlled physical separation	High post-event mass / fire hazard retained with capsule	Release hardware, retention, sequencing and separation dynamics must be demonstrated
S5	Capsule structural + occupant protection	Impact / separation loads	Must not depend on successful jettison; failed / partial separation cases require analysis
S6	Recovery / post-event survival	Descent and post-landing exposure	Parachute, stability, recovery aids and life-support must be sized and verified

Safety-layer rule: S1–S3 are initiation functions, S4 is the separation function, and S5–S6 are residual survival functions. They are deliberately not described as six independent barriers. The safety case must identify dependencies, common-cause failures and conditional probabilities before quantitative credit is assigned.

10. RAPID SEPARATION / JETTISON ARCHITECTURE

- Candidate separable items may include engines, selected fuel sources, the tail-rotor system, selected external structure and selected rotor components only where release is shown to be dynamically and structurally acceptable.
- Main-rotor release is not a default requirement. Any rotor-release concept must address aerodynamic forces, gyroscopic effects, rotor inertia, blade / hub trajectories, tail interaction, capsule clearance and loss of control before separation.
- Every release interface requires a normal-flight load case, emergency release case, retention case, failed-release case and unintended-release case.
- Fuel isolation shall precede or accompany release according to the validated hazard analysis; residual fuel, pressure, ignition sources and line-break behaviour must be controlled.
- Separation-state sensing shall be used where required to prevent an unsafe downstream action (for example parachute deployment into uncleared structure).
- Partial and asymmetric separation are first-class design cases, not afterthoughts.

Phase	Development objective	Engineering condition
1. Command validation	0–0.4 s	Validated command, inhibit checks and safe release-state establishment
2. Primary isolation / release	0.4–2.2 s	Only releases proven safe for the current dynamic state

Phase	Development objective	Engineering condition
3. Secondary release	2.2–3.8 s	Release additional designated items only if trajectory / loads remain acceptable
4. Capsule release	3.8–5.4 s	Release remaining capsule attachments and verify state
5. Capsule-clear decision	5.4–6.0 s	Authorise parachute deployment only if clearance logic is satisfied

The ≤ 6 s sequence is a development objective only. It may be lengthened, restructured or deleted if safety, dynamics, reliability or certification evidence shows that the objective is unsuitable. Configuration control is mandatory.

11. POST-SEPARATION DYNAMICS AND IMPACT PROTECTION

For the 4 500 kg planning capsule mass, $E = \frac{1}{2}mv^2$ yields the following point values for vertical kinetic energy only:

Vertical speed	Energy	Interpretation
6 m/s	81.0 kJ	Point calculation only
8 m/s	144.0 kJ	Point calculation only
10 m/s	225.0 kJ	Point calculation only

- Impact energy alone does not define occupant survivability. Load path, stroke, crush behaviour, deceleration pulse, seat response and human tolerance must be modelled together.
- 6-DOF capsule dynamics shall include translational and rotational states, mass properties, parachute loads, release impulses and component collision envelopes.
- Ground cases shall include vertical, lateral and longitudinal velocities; pitch / roll / yaw rates; slope; uneven terrain; soft soil; water where required; asymmetric leg engagement; and failed-leg cases.
- Acceptance criteria shall be defined by the applicable occupant-protection and structural requirements before compliance testing.

12. PARACHUTE RECOVERY SYSTEM

- The final parachute count shall be an output of sizing and reliability analysis, not a fixed number at this system-baseline stage.
- Design variables include capsule mass, target descent rate, deployment altitude, airspeed, dynamic pressure, temperature, canopy drag, reefing, inflation loads, suspension loads and stability.
- The deployment sequence shall account for verified capsule clearance and shall address partial deployment, single-canopy failure, line / canopy damage, asymmetric inflation and oscillation.
- Parachute reliability shall be allocated quantitatively and verified using appropriate analysis, component tests, deployment tests and system-level demonstrations.
- Parachute deployment into a non-cleared rotor, fuel, tail or structural hazard is a prohibited design condition unless a specific, independently justified fallback mode is demonstrated.

13. STRUCTURAL, MATERIAL AND LOAD-PATH BASELINE

- Normal-flight capsule attachments and emergency-release interfaces shall have separately documented load paths.
- CFRP structure shall address joints, local bearing / bypass, delamination / damage tolerance, environmental degradation, lightning / environmental protection and inspection / repair philosophy.
- Metallic nodes and release fittings shall be assessed for fatigue, ultimate / static strength, thermal / environmental effects and accidental load cases.
- Parachute opening loads, release shock, landing loads and asymmetric separation loads shall be included in the structural design envelope.
- Any interface crossing a safety-layer boundary shall be assessed for common-cause vulnerability and inadvertent energy transfer.

14. AVIONICS, CONTROL LOGIC AND HUMAN FACTORS

- Automatic initiation logic shall define sensor validity, voting / confirmation logic, arming conditions, inhibits, timing, fault handling and failure annunciation.

- Manual initiation shall be accessible to both pilots where applicable, protected against inadvertent actuation and supported by unambiguous system-state indication.
- Emergency logic shall define behaviour under power interruption, sensor disagreement, communication loss, processor reset and partial-system availability.
- Human-factors analysis shall address workload, false activation, confirmation time, cockpit visibility, training assumptions and degraded-mode operation.
- Software and complex electronics shall be developed under the standards and assurance levels required by the applicable certification basis.

15. SAFETY ASSESSMENT AND COMMON-CAUSE ANALYSIS

Analysis	Purpose	Status
FHA	Identify hazards, failure conditions and severity	[OPEN]
PSSA / architecture safety assessment	Allocate safety requirements to architecture	[OPEN]
FMEA / FMECA	Identify component-level failure modes and effects	[OPEN]
Fault trees	Quantify top-event paths and dependencies	[OPEN]
Common-cause analysis	Find shared causes across supposed layers	[OPEN]
Zonal / interface analysis	Control physical proximity and shared hazards	[OPEN]
Human factors	Control crew-induced and interface hazards	[OPEN]
Fire / fuel hazard analysis	Control ignition, leakage, isolation and separation hazards	[OPEN]

16. PRINCIPAL INTERFACES AND ICD CONTROL

Interface	Minimum controlled content
Capsule ↔ airframe	Static / dynamic loads, CG, attachment stiffness, release sequence, clearance
Release mechanism ↔ structure	Retention loads, actuation energy, fatigue, release timing, failure modes
Fuel ↔ separation boundary	Isolation, residual pressure, leakage, fire, line break and ignition control
Electrical / power ↔ emergency system	Independent power, isolation, inhibits, fault containment and EMC
Crew ↔ emergency system	Controls, indication, arming, confirmation, workload and inadvertent-actuation protection
Capsule ↔ parachute	Deployment command, suspension loads, clearance, failure states and stability
Capsule ↔ landing system	Load transfer, stroke, damping law, failed-leg cases and inspection
Aircraft ↔ rotor / transmission	Power, torque, vibration, control, dynamic loads and transient limits

All cross-boundary interfaces shall receive an ICD or equivalent controlled interface definition before detailed design freeze.

17. ENVIRONMENT AND OPERATING ENVELOPE

- Define altitude, temperature, pressure, humidity, wind, precipitation, icing, dust / sand, vibration, shock, lightning and electromagnetic environment.
- Define the emergency initiation and separation envelope in terms of airspeed, altitude, attitude, angular rates, rotor state, engine state, fuel state and configuration.
- Each out-of-envelope emergency case shall be classified as protected, conditionally protected, or unprotected, with an explicit rationale.
- Environmental qualification requirements shall be allocated to the relevant subsystem and verified by the applicable test / analysis methods.

18. MASTER VERIFICATION GATES

Gate	Closure evidence	Status
VG-LSH-01	Complete requirements database and traceability matrix	[OPEN]
VG-LSH-02	Mass, CG, inertia and configuration-state closure	[OPEN]
VG-LSH-03	Propulsion / rotor power and performance closure	[OPEN]

Gate	Closure evidence	Status
VG-LSH-04	Capsule structural / load-path and damage-tolerance basis	[OPEN]
VG-LSH-05	Separation mechanism loads, reliability and timing evidence	[OPEN]
VG-LSH-06	6-DOF separation dynamics and collision analysis	[OPEN]
VG-LSH-07	Parachute sizing, deployment and reliability evidence	[OPEN]
VG-LSH-08	Landing attenuation and occupant-load evidence	[OPEN]
VG-LSH-09	Safety case, FMEA / FMECA, FTA and common-cause closure	[OPEN]
VG-LSH-10	Unintended-jettison probability allocation and evidence	[OPEN]
VG-LSH-11	Certification basis and means-of-compliance mapping	[OPEN]
VG-LSH-12	Integrated ground / flight verification and validation plan	[OPEN]

19. REQUIREMENTS TRACEABILITY AND DOCUMENT TREE RULES

- Every LSH requirement receives a unique identifier, source, rationale, verification method and allocated subsystem.
- Every target and assumption receives an owner, maturity state and closure document.
- Every interface receives an ICD identifier and parent / child ownership.
- Every analysis identifies inputs, assumptions, model version, boundary conditions, uncertainty and acceptance criteria.
- Every verification result points back to the requirement and forward to objective evidence.
- Child documents may refine an allocated requirement but may not silently weaken, contradict or redefine a parent requirement.
- Revision history shall preserve the origin of superseded assumptions and requirements.

Engineering chain	Required relationship
LSH 00	System baseline and top-level requirements
System architecture	Allocated functions, boundaries and safety architecture
Subsystem baselines	Detailed requirements and interfaces
Component / detail design	Geometry, materials, loads and manufacturing definition
Analysis documents	Models, assumptions, margins and predicted behaviour
Verification documents	Test / inspection / analysis procedures and acceptance criteria
Evidence records	Approved results, anomalies, corrective action and closure
Certification package	Means of compliance, conformity and approved evidence

20. CONFIGURATION MANAGEMENT AND EXTERNAL REPRESENTATION

- LSH 00 Rev 3.2 is a proposed root baseline and is not frozen for production.
- Changes to MTOW, capsule mass, rotor diameter, propulsion class, separation logic, parachute architecture or safety claims require impact assessment and configuration control.
- External presentations shall clearly distinguish concept, target, analysis prediction and demonstrated evidence.
- No document derived from LSH 00 may claim certification approval or operational safety unless the applicable authority and evidence support that statement.

21. TECHNOLOGY / LICENSING PACKAGE READINESS

LSH 00 can serve as the engineering root of a future technology-licensing package, but the licenceable asset must be defined as a controlled collection of rights and technical evidence rather than as this single document.

- **Engineering package:** requirements, system architecture, CAD, drawings, analysis models, test procedures / results, manufacturing definition and maintenance information as applicable.
- **IP package:** patents / patent applications, trade secrets, know-how, software / data rights and documented ownership chain.
- **Certification package:** applicable certification basis, compliance matrix, conformity plan and objective evidence.

- **Commercial package:** territory, field of use, exclusivity, improvements, sublicensing, royalties, milestones, support, confidentiality and termination / continuity provisions.
- Jurisdiction-specific sanctions, export-control, technology-transfer and contract restrictions require independent qualified legal review.

22. CRITICAL OPEN ENGINEERING ITEMS

- Bottom-up capsule mass closure and full vehicle mass properties.
- Propulsion / rotor feasibility at the 15 000 kg planning mass and relevant hot / high conditions.
- Definition of the actual thrust / power margin requirement and its operating points.
- Detailed capsule load paths and structural substantiation.
- Release-interface architecture and proof that candidate jettisoned components can separate without unacceptable hazards.
- Post-separation 6-DOF dynamics, including component trajectories and parachute-clearance analysis.
- Parachute sizing, final canopy count, deployment envelope and reliability allocation.
- Landing attenuation design and occupant injury / load criteria.
- Quantitative unintended-jettison probability allocation.
- Applicable certification basis, standards and means of compliance.

23. BASELINE ACCEPTANCE STATEMENT

Engineering disposition: LSH 00 Rev 3.2 is suitable as the root system-level preliminary baseline for controlled decomposition into a large engineering document tree. It is intentionally not a claim that the concept is already feasible, certifiable or safe in operation.

- The previous disk-loading ambiguity has been removed.
- The previous implied propulsion-feasibility claim has been removed.
- The parachute count is no longer prematurely frozen.
- Rotor / component jettison is no longer treated as universally safe or automatically required.
- The six safety functions are now explicitly separated from the claim of independence; independence requires evidence.
- Partial / asymmetric separation, common-cause failures, human factors and downstream state authorisation are explicit design cases.
- Requirements, targets, assumptions, calculations and evidence are traceable by status.
- All critical performance and safety claims remain gated by analysis, test and certification evidence.

24. REFERENCES AND SOURCE BASIS

- Supplied USCC Document 00 Lightweight Jettison Variant, Rev 1.3, as the source concept baseline.
- Supplied engineering-corrected Rev 2.0, Rev 3.0 and Rev 3.1, used as immediate predecessors for this strengthened revision.
- Authoritative certification regulations, approved design standards, test methods and applicable airworthiness requirements remain to be selected and mapped by the responsible certification authority and engineering team.

END OF LSH 00 — LIGHTWEIGHT SAFE HELICOPTER

Generation 1 | Revision 3.2 | Strengthened Preliminary Controlled System Baseline

Not a certification approval | No unverified performance represented as demonstrated